

Maximum Likelihood and Maximum A Posteriori

Theory, motivation, and full step-by-step derivations for every distribution

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1 Why any of this exists

1.1 The problem: we have data, we want a model

Suppose we have data $x_1, \dots, x_n$ that we believe were generated by some probability distribution $p(x \mid \theta)$, where θ is an unknown parameter (or vector of parameters) — a coin’s bias, a Gaussian’s mean and variance, a regression line’s slopes. We never observe θ directly. All we have is the data. The entire question of “fitting a model” is really the question:

Given the data we saw, what is the most reasonable value of θ ?

There are two classical answers to this question: **Maximum Likelihood Estimation (MLE)** and **Maximum A Posteriori estimation (MAP)**. They differ in exactly one ingredient: whether we allow ourselves an opinion about θ *before* looking at the data.

1.2 Maximum Likelihood Estimation (MLE)

Theory — MLE

The **likelihood** of a parameter value θ is the probability the model *would have* assigned to the data we actually saw, treated as a function of θ with the data frozen:

$$L(\theta) = p(x_1, \dots, x_n \mid \theta) \stackrel{\text{i.i.d.}}{=} \prod_{i=1}^n p(x_i \mid \theta).$$

MLE picks the θ that makes this as large as possible:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} p(x \mid \theta).$$

In words: *out of every value θ could take, which one makes today’s data the least surprising?* MLE never asks whether θ itself is plausible on general grounds — it only asks how well θ explains what we observed. It is a purely data-driven estimator.

Why we take logs. Products of n terms are unpleasant to differentiate (product rule blows up combinatorially) and numerically they underflow to zero for even moderate n . Since log is strictly increasing, it does not move the arg max:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \log L(\theta) = \arg \max_{\theta} \sum_{i=1}^n \log p(x_i \mid \theta) =: \arg \max_{\theta} \ell(\theta).$$

$\ell(\theta)$ is called the **log-likelihood**. Because most software minimises rather than maximises, we often flip the sign and define the **negative log-likelihood** $\text{NLL}(\theta) = -\ell(\theta)$, so that $\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} \text{NLL}(\theta)$. This is exactly why “minimise the loss” (squared error, cross-entropy, ...) and “maximise the likelihood” are the same activity wearing different clothes.

1.3 Why MLE is not always enough

Where MLE breaks

Toss a coin twice, see two heads. MLE says $\hat{\theta} = 1$: the coin *never* shows tails, and the model now assigns probability exactly zero to ever seeing a tail again. This is absurd given how little data we have, but MLE has no mechanism to say “that’s an extreme conclusion for only 2 data points” — it only sees what maximises the likelihood of the data in hand. With small n , or with data that happens to be unrepresentative, MLE **overfits**: it is too willing to chase the sample.

The fix is to let ourselves have an opinion about θ *before* seeing any data, and to let the data update that opinion rather than overwrite it completely. That opinion is the **prior**, and updating it with data is exactly **Bayes’ rule**.

1.4 Maximum A Posteriori estimation (MAP)

Theory — MAP

Put a distribution $p(\theta)$ on the parameter itself, representing what we believe about θ *before* seeing y . Bayes’ rule turns this, plus the likelihood, into a distribution over θ *after* seeing the data:

$$p(\theta | x) = \frac{p(x | \theta) p(\theta)}{p(x)}.$$

The four pieces have names:

$p(\theta x)$	posterior — belief about θ after the data
$p(x \theta)$	likelihood — how well this θ explains x
$p(\theta)$	prior — belief about θ before the data
$p(x)$	evidence — a constant with respect to θ

MAP estimation picks the peak of the posterior:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta | x).$$

Why MAP over MLE. MAP is the compromise between what the data wants and what we already believed. Picture three curves: a prior peaked at our prior guess, a likelihood peaked at $\hat{\theta}_{\text{MLE}}$, and a posterior peaked somewhere *between* the two. With very little data the likelihood is wide and uncertain, so the prior dominates and MAP stays close to our prior belief — exactly what protects against the two-heads-in-a-row disaster above. With a lot of data the likelihood becomes sharp and swamps the prior, and MAP is dragged toward MLE. So MAP does not replace MLE; it *subsumes* it as the special case of “no prior opinion.”

Theory — uniform prior implies MAP equals MLE

If $p(\theta)$ is uniform (flat) over all valid θ , then $\log p(\theta)$ is a constant that does not depend on θ , so it cannot move the arg max:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} [\log p(x | \theta) + \underbrace{\log p(\theta)}_{\text{constant}}] = \arg \max_{\theta} \log p(x | \theta) = \hat{\theta}_{\text{MLE}}.$$

A uniform prior is, formally, “I refuse to have an opinion.” It is the honest way to say MAP reduces to MLE.

1.5 The MAP derivation, in general (the master recipe)

Derivation — general MAP recipe

$$\begin{aligned} \hat{\theta}_{\text{MAP}} &= \arg \max_{\theta} p(\theta | x) = \arg \max_{\theta} \log p(\theta | x) \\ &= \arg \max_{\theta} \log \frac{p(x | \theta) p(\theta)}{p(x)} \\ &= \arg \max_{\theta} [\log p(x | \theta) + \log p(\theta) - \log p(x)]. \end{aligned}$$

Drop $\log p(x)$: it has no θ in it. Expand the likelihood using independence:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \left[\sum_{i=1}^n \log p(x_i | \theta) + \log p(\theta) \right] = \arg \max_{\theta} [\ell(\theta) + \log p(\theta)].$$

Flip to a minimisation, matching the NLL convention:

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} [\text{NLL}(\theta) - \log p(\theta)].$$

The extra term $-\log p(\theta)$ is exactly what machine learning calls a **regularizer**. This single equation is the parent of every derivation in this document: every distribution below is this same recipe, with $p(x | \theta)$ and $p(\theta)$ swapped in.

Why we always use the *conjugate* prior below. For a family like Beta, Gamma, Gaussian, Dirichlet, etc., there is a special choice of prior — the **conjugate prior** — for which the posterior $p(\theta | x)$ belongs to the *same family* as the prior. This is what makes the algebra below close in a few lines instead of requiring numerical integration; it is also historically why these particular prior/likelihood pairings are the “standard” examples taught together (Beta–Bernoulli, Gamma–Poisson, Gaussian–Gaussian, and so on).

2 Roadmap

For each distribution below we derive, *fully*, in the same seven-part shape:

1. State the model $p(x \mid \theta)$, and for MAP, the conjugate prior $p(\theta)$.
2. Write the log-likelihood $\ell(\theta) = \sum_i \log p(x_i \mid \theta)$, expanding every term.
3. (MAP only) Write the log-prior $\log p(\theta)$, expanding every term.
4. Add log-likelihood (+ log-prior for MAP), drop every additive constant that has no θ .
5. Differentiate with respect to the parameter.
6. Set the derivative to zero and solve algebraically, showing every line.
7. Box the estimator, and sanity-check it against intuition.

3 Bernoulli(θ)

$$p(x | \theta) = \theta^x (1 - \theta)^{1-x}, \quad x \in \{0, 1\}.$$

Check: if $x = 1$ this is θ ; if $x = 0$ this is $1 - \theta$. Good, matches “probability of heads is θ .”

3.1 MLE

Derivation

Step 1 — log-likelihood. With $s = \sum_i x_i$ (number of heads):

$$\ell(\theta) = \sum_{i=1}^n \log [\theta^{x_i} (1 - \theta)^{1-x_i}] = \sum_{i=1}^n [x_i \log \theta + (1 - x_i) \log(1 - \theta)].$$

Step 2 — collect terms. $\sum_i x_i = s$ and $\sum_i (1 - x_i) = n - s$:

$$\ell(\theta) = s \log \theta + (n - s) \log(1 - \theta).$$

Step 3 — differentiate. Using $\frac{d}{d\theta} \log \theta = 1/\theta$ and $\frac{d}{d\theta} \log(1 - \theta) = -1/(1 - \theta)$:

$$\frac{d\ell}{d\theta} = \frac{s}{\theta} - \frac{n - s}{1 - \theta}.$$

Step 4 — set to zero and clear denominators. Multiply both sides by $\theta(1 - \theta)$:

$$s(1 - \theta) - (n - s)\theta = 0 \implies s - s\theta - n\theta + s\theta = 0 \implies s - n\theta = 0.$$

Step 5 — solve.

$$\hat{\theta}_{\text{MLE}} = \frac{s}{n}$$

Sanity check. 7 heads in 10 tosses $\Rightarrow \hat{\theta} = 0.7$ — exactly the fraction of heads, as intuition demands.

3.2 MAP with a Beta(α, β) prior

Derivation

Step 1 — prior. The Beta density is the conjugate prior on $(0, 1)$:

$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1} \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1},$$

where $B(\alpha, \beta)$ is a normalising constant with no θ in it, so we drop it.

Step 2 — log-prior.

$$\log p(\theta) = (\alpha - 1) \log \theta + (\beta - 1) \log(1 - \theta) + \text{const.}$$

Step 3 — add log-likelihood and log-prior.

$$\ell(\theta) + \log p(\theta) = [s \log \theta + (n - s) \log(1 - \theta)] + [(\alpha - 1) \log \theta + (\beta - 1) \log(1 - \theta)] + \text{const.}$$

Step 4 — collect like terms.

$$= (s + \alpha - 1) \log \theta + (n - s + \beta - 1) \log(1 - \theta) + \text{const.}$$

Step 5 — differentiate.

$$\frac{d}{d\theta} [\ell + \log p] = \frac{s + \alpha - 1}{\theta} - \frac{n - s + \beta - 1}{1 - \theta}.$$

Step 6 — set to zero, clear denominators.

$$(s + \alpha - 1)(1 - \theta) - (n - s + \beta - 1)\theta = 0$$

$$(s + \alpha - 1) - (s + \alpha - 1)\theta - (n - s + \beta - 1)\theta = 0$$

$$(s + \alpha - 1) = \theta[(s + \alpha - 1) + (n - s + \beta - 1)] = \theta(n + \alpha + \beta - 2).$$

Step 7 — solve.

$$\hat{\theta}_{\text{MAP}} = \frac{s + \alpha - 1}{n + \alpha + \beta - 2}$$

Sanity check. $\alpha = \beta = 1$ (Beta(1, 1) is the uniform prior) gives $\hat{\theta}_{\text{MAP}} = s/n = \hat{\theta}_{\text{MLE}}$, matching the “uniform prior = no opinion” theory box. α, β act like “phantom” prior heads/tails added to the count before dividing.

4 Binomial(m, θ)

$$p(k \mid \theta) = \binom{m}{k} \theta^k (1 - \theta)^{m-k}, \quad k \in \{0, \dots, m\}.$$

n independent repeats of “ m tosses,” k_i successes in repeat i , $s = \sum_i k_i$.

4.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(\theta) = \sum_{i=1}^n \log \left[\binom{m}{k_i} \theta^{k_i} (1 - \theta)^{m-k_i} \right] = \sum_{i=1}^n \left[\log \binom{m}{k_i} + k_i \log \theta + (m - k_i) \log(1 - \theta) \right].$$

Step 2 — drop the θ -free term and collect. $\log \binom{m}{k_i}$ has no θ , drop it. With $s = \sum_i k_i$ and $\sum_i (m - k_i) = nm - s$:

$$\ell(\theta) = s \log \theta + (nm - s) \log(1 - \theta) + \text{const.}$$

Step 3 — differentiate, set to zero. Identical algebra to Bernoulli with n replaced by nm :

$$\frac{s}{\theta} - \frac{nm - s}{1 - \theta} = 0 \implies s - nm\theta = 0.$$

Step 4 — solve.

$$\hat{\theta}_{\text{MLE}} = \frac{s}{nm} = \frac{\sum_i k_i}{nm}$$

4.2 MAP with a Beta(α, β) prior

Derivation

Step 1–4 — same algebra as Bernoulli MAP, with $n \rightarrow nm$ everywhere (the log-prior is unchanged, since it lives on θ directly, not on the counts):

$$(s + \alpha - 1) \log \theta + (nm - s + \beta - 1) \log(1 - \theta) + \text{const.}$$

Step 5 — differentiate, set to zero, solve (same steps as Bernoulli MAP):

$$(s + \alpha - 1) = \theta(nm + \alpha + \beta - 2).$$

$$\hat{\theta}_{\text{MAP}} = \frac{s + \alpha - 1}{nm + \alpha + \beta - 2} = \frac{\sum_i k_i + \alpha - 1}{nm + \alpha + \beta - 2}$$

5 Categorical($p_1, \dots, p_K$)

$$p(x | p) = \prod_{k=1}^K p_k^{\mathbb{1}[x=k]}, \quad \sum_{k=1}^K p_k = 1, \quad p_k \geq 0.$$

n_k = count of category k across n samples, $N = \sum_k n_k$.

5.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(p) = \sum_{i=1}^n \sum_{k=1}^K \mathbb{1}[x_i = k] \log p_k = \sum_{k=1}^K n_k \log p_k.$$

Step 2 — this is a constrained maximisation ($\sum_k p_k = 1$), so plain differentiation is not enough; introduce a Lagrange multiplier λ :

$$\mathcal{L}(p, \lambda) = \sum_{k=1}^K n_k \log p_k + \lambda \left(1 - \sum_{k=1}^K p_k \right).$$

Step 3 — differentiate with respect to each p_k , set to zero.

$$\frac{\partial \mathcal{L}}{\partial p_k} = \frac{n_k}{p_k} - \lambda = 0 \implies p_k = \frac{n_k}{\lambda} \quad \text{for every } k = 1, \dots, K.$$

Step 4 — differentiate with respect to λ , set to zero (recover the constraint).

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - \sum_k p_k = 0 \implies \sum_k p_k = 1.$$

Step 5 — substitute Step 3 into Step 4 to solve for λ .

$$\sum_{k=1}^K \frac{n_k}{\lambda} = 1 \implies \frac{1}{\lambda} \sum_k n_k = 1 \implies \frac{N}{\lambda} = 1 \implies \lambda = N.$$

Step 6 — back-substitute $\lambda = N$ into Step 3.

$$\hat{p}_k = \frac{n_k}{N}$$

5.2 MAP with a Dirichlet($\alpha_1, \dots, \alpha_K$) prior

Derivation

Step 1 — prior.

$$p(p) \propto \prod_{k=1}^K p_k^{\alpha_k - 1}, \quad \log p(p) = \sum_{k=1}^K (\alpha_k - 1) \log p_k + \text{const.}$$

Step 2 — Lagrangian of log-likelihood + log-prior.

$$\mathcal{L}(p, \lambda) = \sum_{k=1}^K (n_k + \alpha_k - 1) \log p_k + \lambda \left(1 - \sum_k p_k \right).$$

Step 3 — differentiate w.r.t. p_k , set to zero.

$$\frac{n_k + \alpha_k - 1}{p_k} - \lambda = 0 \implies p_k = \frac{n_k + \alpha_k - 1}{\lambda}.$$

Step 4 — apply the constraint $\sum_k p_k = 1$:

$$\sum_{k=1}^K \frac{n_k + \alpha_k - 1}{\lambda} = 1 \implies \lambda = \sum_{k=1}^K (n_k + \alpha_k - 1) = N + \sum_{k=1}^K (\alpha_k - 1).$$

Step 5 — solve.

$$\hat{p}_k^{\text{MAP}} = \frac{n_k + \alpha_k - 1}{N + \sum_{j=1}^K (\alpha_j - 1)}$$

Sanity check. Each $\alpha_k - 1$ acts as a “phantom count” added to the observed count n_k before normalising — the multi-category analogue of the Beta–Bernoulli pseudo-counts.

6 Poisson(λ)

$$p(k \mid \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

6.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(\lambda) = \sum_{i=1}^n \log \left[\frac{\lambda^{k_i} e^{-\lambda}}{k_i!} \right] = \sum_{i=1}^n [k_i \log \lambda - \lambda - \log k_i!].$$

Step 2 — drop the λ -free term, collect. $\log k_i!$ has no λ :

$$\ell(\lambda) = \left(\sum_i k_i \right) \log \lambda - n\lambda + \text{const.}$$

Step 3 — differentiate.

$$\frac{d\ell}{d\lambda} = \frac{\sum_i k_i}{\lambda} - n.$$

Step 4 — set to zero, solve.

$$\frac{\sum_i k_i}{\lambda} = n \implies \boxed{\hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n k_i = \bar{k}}$$

6.2 MAP with a Gamma(a, b) prior

Derivation

Step 1 — prior. Gamma(a, b) with shape a , rate b :

$$p(\lambda) \propto \lambda^{a-1} e^{-b\lambda}, \quad \log p(\lambda) = (a-1) \log \lambda - b\lambda + \text{const.}$$

Step 2 — add to log-likelihood.

$$\ell(\lambda) + \log p(\lambda) = \left(\sum_i k_i \right) \log \lambda - n\lambda + (a-1) \log \lambda - b\lambda + \text{const.}$$

Step 3 — collect like terms.

$$= \left(\sum_i k_i + a - 1 \right) \log \lambda - (n + b)\lambda + \text{const.}$$

Step 4 — differentiate.

$$\frac{d}{d\lambda} [\ell + \log p] = \frac{\sum_i k_i + a - 1}{\lambda} - (n + b).$$

Step 5 — set to zero, solve.

$$\frac{\sum_i k_i + a - 1}{\lambda} = n + b \implies \boxed{\hat{\lambda}_{\text{MAP}} = \frac{\sum_i k_i + a - 1}{n + b}}$$

Sanity check. $a - 1$ pseudo-events and b pseudo-trials are added to the numerator and denominator of the sample mean.

7 Geometric(p) — number of trials to first success

$$P(k | p) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

7.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(p) = \sum_{i=1}^n \log [(1 - p)^{k_i-1}p] = \sum_{i=1}^n [(k_i - 1) \log(1 - p) + \log p].$$

Step 2 — collect terms.

$$\ell(p) = \left(\sum_i k_i - n \right) \log(1 - p) + n \log p.$$

Step 3 — differentiate. $\frac{d}{dp} \log p = 1/p$, $\frac{d}{dp} \log(1 - p) = -1/(1 - p)$:

$$\frac{d\ell}{dp} = \frac{n}{p} - \frac{\sum_i k_i - n}{1 - p}.$$

Step 4 — set to zero, clear denominators.

$$n(1 - p) - \left(\sum_i k_i - n \right) p = 0 \implies n - np - p \sum_i k_i + np = 0 \implies n = p \sum_i k_i.$$

Step 5 — solve.

$$\hat{p}_{\text{MLE}} = \frac{n}{\sum_i k_i} = \frac{1}{\bar{k}}$$

Sanity check. If on average it takes 4 tosses to see a head, $\hat{p} = 1/4$ — matches intuition for a fair-ish coin.

7.2 MAP with a Beta(α, β) prior

Derivation

Step 1 — log-prior.

$$\log p(p) = (\alpha - 1) \log p + (\beta - 1) \log(1 - p) + \text{const.}$$

Step 2 — add to log-likelihood and collect.

$$\ell(p) + \log p(p) = (n + \alpha - 1) \log p + \left(\sum_i k_i - n + \beta - 1 \right) \log(1 - p) + \text{const.}$$

Step 3 — differentiate.

$$\frac{n + \alpha - 1}{p} - \frac{\sum_i k_i - n + \beta - 1}{1 - p}.$$

Step 4 — set to zero, clear denominators, solve (same mechanics as Bernoulli MAP):

$$(n + \alpha - 1)(1 - p) = \left(\sum_i k_i - n + \beta - 1 \right) p$$

$$(n + \alpha - 1) = p \left[(n + \alpha - 1) + \left(\sum_i k_i - n + \beta - 1 \right) \right] = p \left(\sum_i k_i + \alpha + \beta - 2 \right).$$

$$\hat{p}_{\text{MAP}} = \frac{n + \alpha - 1}{\sum_i k_i + \alpha + \beta - 2}$$

8 Gaussian mean, variance σ^2 known

$$p(x \mid \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

8.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(\mu) = \sum_{i=1}^n \left[-\log \sqrt{2\pi\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right] = -n \log \sqrt{2\pi\sigma^2} - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Step 2 — drop the μ -free term. $-n \log \sqrt{2\pi\sigma^2}$ has no μ :

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 + \text{const.}$$

Step 3 — differentiate. Chain rule on $(x_i - \mu)^2$: derivative w.r.t. μ is $-2(x_i - \mu)$:

$$\frac{d\ell}{d\mu} = -\frac{1}{2\sigma^2} \sum_i 2(x_i - \mu)(-1) = \frac{1}{\sigma^2} \sum_i (x_i - \mu).$$

Step 4 — set to zero.

$$\sum_i (x_i - \mu) = 0 \implies \sum_i x_i - n\mu = 0.$$

Step 5 — solve.

$$\hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_i x_i = \bar{x}$$

8.2 MAP with a Gaussian prior $\mu \sim \mathcal{N}(\mu_0, \tau^2)$

Derivation

Step 1 — log-prior.

$$\log p(\mu) = -\log \sqrt{2\pi\tau^2} - \frac{(\mu - \mu_0)^2}{2\tau^2} = -\frac{(\mu - \mu_0)^2}{2\tau^2} + \text{const.}$$

Step 2 — add to log-likelihood.

$$\ell(\mu) + \log p(\mu) = -\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 - \frac{1}{2\tau^2} (\mu - \mu_0)^2 + \text{const.}$$

Step 3 — differentiate.

$$\frac{d}{d\mu} [\ell + \log p] = \frac{1}{\sigma^2} \sum_i (x_i - \mu) - \frac{1}{\tau^2} (\mu - \mu_0).$$

Step 4 — set to zero.

$$\frac{1}{\sigma^2} \sum_i (x_i - \mu) - \frac{1}{\tau^2} (\mu - \mu_0) = 0.$$

Step 5 — expand $\sum_i (x_i - \mu) = n\bar{x} - n\mu$.

$$\frac{n\bar{x} - n\mu}{\sigma^2} - \frac{\mu - \mu_0}{\tau^2} = 0.$$

Step 6 — multiply through by $\sigma^2\tau^2$ to clear denominators.

$$\tau^2(n\bar{x} - n\mu) - \sigma^2(\mu - \mu_0) = 0 \implies n\tau^2\bar{x} - n\tau^2\mu - \sigma^2\mu + \sigma^2\mu_0 = 0.$$

Step 7 — collect the μ terms on one side.

$$\mu(n\tau^2 + \sigma^2) = n\tau^2\bar{x} + \sigma^2\mu_0.$$

Step 8 — solve.

$$\hat{\mu}_{\text{MAP}} = \frac{n\tau^2\bar{x} + \sigma^2\mu_0}{n\tau^2 + \sigma^2}$$

Sanity check. This is a weighted average of the data mean $\bar{x}$ and the prior mean μ_0 , weighted by their precisions n/σ^2 and $1/\tau^2$. Divide numerator and denominator by $\sigma^2\tau^2$ to see it explicitly:

$$\hat{\mu}_{\text{MAP}} = \frac{\frac{n}{\sigma^2}\bar{x} + \frac{1}{\tau^2}\mu_0}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}.$$

As $n \rightarrow \infty$ the data term dominates and $\hat{\mu}_{\text{MAP}} \rightarrow \bar{x} = \hat{\mu}_{\text{MLE}}$ — more data drags MAP toward MLE, exactly the “black curve gets sharper” picture from the theory section.

9 Gaussian variance, mean μ known

Let $v = \sigma^2$ to keep the algebra clean (we are estimating the *variance*, so differentiate with respect to v , not σ).

9.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(v) = \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi v) - \frac{(x_i - \mu)^2}{2v} \right] = -\frac{n}{2} \log(2\pi v) - \frac{1}{2v} \sum_i (x_i - \mu)^2.$$

Step 2 — split $\log(2\pi v) = \log(2\pi) + \log v$ and drop the v -free piece.

$$\ell(v) = -\frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \mu)^2 + \text{const.}$$

Step 3 — differentiate. Using $\frac{d}{dv} \log v = 1/v$ and $\frac{d}{dv} v^{-1} = -v^{-2}$:

$$\frac{d\ell}{dv} = -\frac{n}{2v} + \frac{1}{2v^2} \sum_i (x_i - \mu)^2.$$

Step 4 — set to zero, multiply through by $2v^2$.

$$-nv + \sum_i (x_i - \mu)^2 = 0 \implies \sum_i (x_i - \mu)^2 = nv.$$

Step 5 — solve.

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_i (x_i - \mu)^2$$

9.2 MAP with an Inverse-Gamma(a, b) prior

Derivation

Step 1 — prior. Inverse-Gamma is the conjugate prior for a Gaussian variance:

$$p(v) \propto v^{-(a+1)} e^{-b/v}, \quad \log p(v) = -(a+1) \log v - \frac{b}{v} + \text{const.}$$

Step 2 — add to log-likelihood.

$$\ell(v) + \log p(v) = -\frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \mu)^2 - (a+1) \log v - \frac{b}{v} + \text{const.}$$

Step 3 — collect $\log v$ terms and $1/v$ terms separately.

$$= -\left(\frac{n}{2} + a + 1\right) \log v - \frac{1}{v} \left[\frac{1}{2} \sum_i (x_i - \mu)^2 + b \right] + \text{const.}$$

Step 4 — differentiate.

$$\frac{d}{dv} [\ell + \log p] = -\frac{\frac{n}{2} + a + 1}{v} + \frac{\frac{1}{2} \sum_i (x_i - \mu)^2 + b}{v^2}.$$

Step 5 — set to zero, multiply through by v^2 .

$$-\left(\frac{n}{2} + a + 1\right)v + \left[\frac{1}{2} \sum_i (x_i - \mu)^2 + b\right] = 0.$$

Step 6 — solve.

$$\hat{\sigma}_{\text{MAP}}^2 = \frac{\frac{1}{2} \sum_i (x_i - \mu)^2 + b}{\frac{n}{2} + a + 1}$$

Sanity check. b is a pseudo sum-of-squares and $a + 1$ contributes pseudo-observations, both added before dividing — the Inverse-Gamma prior effectively injects “ $2a + 2$ prior data points worth $2b$ of squared error.”

10 Gaussian, both mean and variance unknown

10.1 MLE — a two-pass argument

Derivation

Step 1 — write the joint log-likelihood.

$$\ell(\mu, v) = -\frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \mu)^2 + \text{const.}$$

Step 2 — differentiate with respect to μ first, holding v fixed. This is Section 7's Step 3 verbatim:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v} \sum_i (x_i - \mu).$$

Step 3 — set to zero. Notice v cancels out — this equation is independent of v :

$$\sum_i (x_i - \mu) = 0 \implies \boxed{\hat{\mu}_{\text{MLE}} = \bar{x}} \quad (\text{true for any value of } v).$$

Step 4 — substitute $\hat{\mu} = \bar{x}$ back into $\ell(\mu, v)$, reducing it to a function of v alone:

$$\ell(\bar{x}, v) = -\frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \bar{x})^2 + \text{const.}$$

Step 5 — differentiate with respect to v (identical shape to Section 7's Step 3, with $\mu \rightarrow \bar{x}$):

$$\frac{d}{dv} \ell(\bar{x}, v) = -\frac{n}{2v} + \frac{1}{2v^2} \sum_i (x_i - \bar{x})^2.$$

Step 6 — set to zero, solve.

$$\sum_i (x_i - \bar{x})^2 = nv \implies \boxed{\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2}$$

Why the two-pass trick is valid. A joint maximum of $\ell(\mu, v)$ requires *both* partial derivatives to vanish simultaneously. Step 3 shows $\partial \ell / \partial \mu = 0$ forces $\mu = \bar{x}$ **no matter what v is**. So plugging $\mu = \bar{x}$ into the equation for v before solving loses nothing — we would have reached the identical pair $(\bar{x}, \frac{1}{n} \sum (x_i - \bar{x})^2)$ by solving both equations together.

10.2 MAP with a Normal-Inverse-Gamma prior

The conjugate prior for the joint case ties the two parameters together: $\mu \mid v \sim \mathcal{N}(\mu_0, v/\kappa_0)$ and $v \sim \text{Inverse-Gamma}(a_0, b_0)$.

Derivation

Step 1 — prior density (dropping normalising constants).

$$p(\mu, v) \propto v^{-1/2} \exp\left(-\frac{\kappa_0(\mu - \mu_0)^2}{2v}\right) \cdot v^{-(a_0+1)} \exp\left(-\frac{b_0}{v}\right).$$

Step 2 — log-prior.

$$\log p(\mu, v) = -\left(a_0 + \frac{3}{2}\right) \log v - \frac{\kappa_0(\mu - \mu_0)^2}{2v} - \frac{b_0}{v} + \text{const.}$$

Step 3 — log-posterior (add to the log-likelihood from the box above):

$$-\frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \mu)^2 - \left(a_0 + \frac{3}{2}\right) \log v - \frac{\kappa_0(\mu - \mu_0)^2}{2v} - \frac{b_0}{v}.$$

Step 4 — differentiate with respect to μ first (only the 2nd and 4th terms involve μ):

$$\frac{\partial}{\partial \mu} = \frac{1}{v} \sum_i (x_i - \mu) - \frac{\kappa_0}{v} (\mu - \mu_0).$$

Step 5 — set to zero; the $1/v$ cancels out of every term.

$$\sum_i (x_i - \mu) - \kappa_0(\mu - \mu_0) = 0 \implies n\bar{x} - n\mu - \kappa_0\mu + \kappa_0\mu_0 = 0.$$

Step 6 — collect μ terms, solve.

$$\mu(n + \kappa_0) = n\bar{x} + \kappa_0\mu_0 \implies \hat{\mu}_{\text{MAP}} = \frac{n\bar{x} + \kappa_0\mu_0}{n + \kappa_0}$$

Step 7 — substitute $\hat{\mu}_{\text{MAP}}$ back and differentiate with respect to v , exactly as in the two-pass MLE argument. The sum of squares now picks up an extra prior-pull term:

$$\text{let } Q = \sum_i (x_i - \hat{\mu}_{\text{MAP}})^2 + \kappa_0(\hat{\mu}_{\text{MAP}} - \mu_0)^2.$$

$$\frac{\partial}{\partial v} \left[-\left(\frac{n}{2} + a_0 + \frac{3}{2}\right) \log v - \frac{Q/2 + b_0}{v} \right] = -\frac{\frac{n}{2} + a_0 + \frac{3}{2}}{v} + \frac{Q/2 + b_0}{v^2}.$$

Step 8 — set to zero, solve.

$$\hat{\sigma}_{\text{MAP}}^2 = \frac{\frac{1}{2} \left[\sum_i (x_i - \hat{\mu}_{\text{MAP}})^2 + \kappa_0(\hat{\mu}_{\text{MAP}} - \mu_0)^2 \right] + b_0}{\frac{n}{2} + a_0 + \frac{3}{2}}$$

11 Exponential(λ)

$$p(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

11.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(\lambda) = \sum_{i=1}^n [\log \lambda - \lambda x_i] = n \log \lambda - \lambda \sum_i x_i.$$

Step 2 — differentiate.

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_i x_i.$$

Step 3 — set to zero, solve.

$$\frac{n}{\lambda} = \sum_i x_i \implies \hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{x}}$$

11.2 MAP with a Gamma(a, b) prior

Derivation

Step 1 — log-prior.

$$\log p(\lambda) = (a - 1) \log \lambda - b\lambda + \text{const.}$$

Step 2 — add to log-likelihood, collect terms.

$$\ell(\lambda) + \log p(\lambda) = (n + a - 1) \log \lambda - \left(\sum_i x_i + b \right) \lambda + \text{const.}$$

Step 3 — differentiate.

$$\frac{n + a - 1}{\lambda} - \left(\sum_i x_i + b \right).$$

Step 4 — set to zero, solve.

$$\hat{\lambda}_{\text{MAP}} = \frac{n + a - 1}{\sum_i x_i + b}$$

Sanity check. Same Gamma–Poisson shape of update, because both likelihoods are exponential-family with a rate parameter — $a - 1$ pseudo-events over b pseudo-time-units are added before dividing.

12 Uniform(0, θ)

$$p(x | \theta) = \frac{1}{\theta}, \quad 0 \leq x \leq \theta.$$

This case is qualitatively different: the density does not depend on x inside its support, but the *support itself* depends on θ . Ordinary calculus (derivative = 0) will not find the maximiser here — the likelihood is monotonic in θ , so the maximum sits at a *boundary*, not an interior stationary point.

12.1 MLE

Derivation

Step 1 — write the full likelihood. Each factor is $1/\theta$, but *only if* $\theta \geq x_i$; otherwise $p(x_i | \theta) = 0$. So:

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} \cdot \mathbb{I}[\theta \geq x_i] = \theta^{-n} \cdot \mathbb{I}[\theta \geq \max_i x_i].$$

Step 2 — note θ^{-n} is strictly decreasing in θ for $\theta > 0$ (its derivative is $-n\theta^{-n-1} < 0$). So among all θ satisfying the indicator, $L(\theta)$ is largest at the *smallest* allowed θ .

Step 3 — the smallest θ satisfying $\theta \geq \max_i x_i$ is $\theta = \max_i x_i$ itself.

$$\hat{\theta}_{\text{MLE}} = \max_i x_i$$

Sanity check. If θ were any smaller than the largest observed data point, that data point would have zero probability under the model — impossible, since we observed it. So θ can never be estimated below $\max_i x_i$; and since larger θ only spreads probability thinner, the smallest feasible value is best.

12.2 MAP with a Pareto(θ_0, a) prior

Derivation

Step 1 — prior. Pareto is the conjugate prior for a Uniform’s upper bound:

$$p(\theta) = \frac{a \theta_0^a}{\theta^{a+1}}, \quad \theta \geq \theta_0.$$

Step 2 — posterior $\propto$ likelihood $\times$ prior:

$$p(\theta | x) \propto \theta^{-n} \cdot \theta^{-(a+1)} \cdot \mathbb{I}[\theta \geq \max_i x_i] \cdot \mathbb{I}[\theta \geq \theta_0] = \theta^{-(n+a+1)} \cdot \mathbb{I}[\theta \geq \max(\max_i x_i, \theta_0)].$$

Step 3 — $\theta^{-(n+a+1)}$ is still strictly decreasing (exponent is negative), so again the maximiser is the smallest feasible θ .

Step 4 — read off the boundary.

$$\hat{\theta}_{\text{MAP}} = \max(\max_i x_i, \theta_0)$$

Sanity check. If the prior’s floor θ_0 is larger than anything observed, MAP trusts the prior and reports θ_0 ; once enough data pushes $\max_i x_i$ above θ_0 , MAP switches to following the data — same “compromise, weighted by which signal is stronger” story as every other MAP case, just expressed through a max instead of a weighted average.

13 Laplace(μ, b), scale b known

$$p(x \mid \mu) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right).$$

13.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(\mu) = \sum_{i=1}^n \left[-\log(2b) - \frac{|x_i - \mu|}{b} \right] = -n \log(2b) - \frac{1}{b} \sum_i |x_i - \mu|.$$

Step 2 — maximising ℓ is the same as minimising $\sum_i |x_i - \mu|$, the L^1 loss. This function is not differentiable exactly where $\mu = x_i$, so we cannot blindly set a derivative to zero everywhere — instead examine the derivative on either side of the data points.

Step 3 — for μ not equal to any x_i ,

$$\frac{d}{d\mu} |x_i - \mu| = \begin{cases} -1 & \mu < x_i \\ +1 & \mu > x_i \end{cases} \implies \frac{d}{d\mu} \sum_i |x_i - \mu| = \#\{i : x_i < \mu\} - \#\{i : x_i > \mu\}.$$

Step 4 — set this to zero: we need the number of data points below μ to equal the number above μ . That is precisely the definition of the median.

$$\boxed{\hat{\mu}_{\text{MLE}} = \text{median}(x_1, \dots, x_n)}$$

Sanity check. Unlike the Gaussian, whose MLE (the mean) is dragged by outliers, the Laplace's heavier tails make the median the natural fit — this is the textbook reason the median is called “robust.”

13.2 MAP with a Laplace(μ_0, c) prior

Derivation**Step 1 — log-prior.**

$$\log p(\mu) = -\log(2c) - \frac{|\mu - \mu_0|}{c} = -\frac{|\mu - \mu_0|}{c} + \text{const.}$$

Step 2 — add to log-likelihood; maximising is minimising a sum of $n + 1$ absolute values, the n data points weighted by $1/b$ and the prior mean μ_0 weighted by $1/c$:

$$\text{minimise } \frac{1}{b} \sum_{i=1}^n |x_i - \mu| + \frac{1}{c} |\mu - \mu_0| \quad \text{over } \mu.$$

Step 3 — the same sign-counting argument as MLE Step 3 applies, but now every “vote” carries a weight ($1/b$ for each x_i , $1/c$ for μ_0): the derivative changes sign at the point where the cumulative weight below equals the cumulative weight above.

$$\hat{\mu}_{\text{MAP}} = \text{weighted median of } \{x_1, \dots, x_n, \mu_0\} \text{ with weights } \frac{1}{b}, \dots, \frac{1}{b}, \frac{1}{c}$$

Sanity check / contrast with every other case in this document. This is the one distribution where MAP is a weighted *median*, not a weighted *mean* — because the loss is L^1 (absolute error), not L^2 (squared error, which is what every Gaussian-flavoured case above reduces to). There is generally no simpler closed form than “sort and find the weighted middle.”

14 Rayleigh(σ)

$$p(x | \sigma) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad x \geq 0.$$

As with the Gaussian variance case, let $v = \sigma^2$.

14.1 MLE

Derivation

Step 1 — log-likelihood.

$$\ell(v) = \sum_{i=1}^n \left[\log x_i - \log v - \frac{x_i^2}{2v} \right] = -n \log v - \frac{1}{2v} \sum_i x_i^2 + \text{const},$$

where $\sum_i \log x_i$ has no v and is dropped as a constant.

Step 2 — differentiate.

$$\frac{d\ell}{dv} = -\frac{n}{v} + \frac{1}{2v^2} \sum_i x_i^2.$$

Step 3 — set to zero, multiply through by $2v^2$.

$$-2nv + \sum_i x_i^2 = 0 \implies \sum_i x_i^2 = 2nv.$$

Step 4 — solve.

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{2n} \sum_i x_i^2$$

14.2 MAP with an Inverse-Gamma(a, b) prior

Derivation

Step 1 — log-prior.

$$\log p(v) = -(a+1) \log v - \frac{b}{v} + \text{const}.$$

Step 2 — add to log-likelihood, collect $\log v$ terms and $1/v$ terms.

$$\ell(v) + \log p(v) = -(n+a+1) \log v - \frac{1}{v} \left[\frac{1}{2} \sum_i x_i^2 + b \right] + \text{const}.$$

Step 3 — differentiate.

$$\frac{d}{dv} [\ell + \log p] = -\frac{n+a+1}{v} + \frac{\frac{1}{2} \sum_i x_i^2 + b}{v^2}.$$

Step 4 — set to zero, multiply through by v^2 , solve.

$$-(n+a+1)v + \left(\frac{1}{2} \sum_i x_i^2 + b \right) = 0.$$

$$\hat{\sigma}_{\text{MAP}}^2 = \frac{\frac{1}{2} \sum_i x_i^2 + b}{n+a+1}$$

15 Master summary table

Distribution	$\hat{\theta}_{\text{MLE}}$	Conjugate prior	$\hat{\theta}_{\text{MAP}}$
Bernoulli(θ)	s/n	Beta(α, β)	$\frac{s + \alpha - 1}{n + \alpha + \beta - 2}$
Binomial(m, θ)	$\frac{\sum_i k_i}{nm}$	Beta(α, β)	$\frac{\sum_i k_i + \alpha - 1}{nm + \alpha + \beta - 2}$
Categorical	n_k/N	Dirichlet(α_k)	$\frac{n_k + \alpha_k - 1}{N + \sum_j (\alpha_j - 1)}$
Poisson(λ)	$\bar{k}$	Gamma(a, b)	$\frac{\sum_i k_i + a - 1}{n + b}$
Geometric(p)	$1/\bar{k}$	Beta(α, β)	$\frac{n + \alpha - 1}{\sum_i k_i + \alpha + \beta - 2}$
Gaussian, mean	$\bar{x}$	$\mathcal{N}(\mu_0, \tau^2)$	$\frac{n\tau^2\bar{x} + \sigma^2\mu_0}{n\tau^2 + \sigma^2}$
Gaussian, var	$\frac{1}{n} \sum_i (x_i - \mu)^2$	Inv-Gamma(a, b)	$\frac{\frac{1}{2} \sum_i (x_i - \mu)^2 + b}{n/2 + a + 1}$
Gaussian, both	$\bar{x}, \frac{1}{n} \sum_i (x_i - \bar{x})^2$	Normal-Inv-Gamma	see §9
Exponential(λ)	$1/\bar{x}$	Gamma(a, b)	$\frac{n + a - 1}{\sum_i x_i + b}$
Uniform($0, \theta$)	$\max_i x_i$	Pareto(θ_0, a)	$\max(\max_i x_i, \theta_0)$
Laplace(μ, b)	$\text{median}(x_i)$	Laplace(μ_0, c)	weighted median
Rayleigh(σ)	$\frac{1}{2n} \sum_i x_i^2$	Inv-Gamma(a, b)	$\frac{\frac{1}{2} \sum_i x_i^2 + b}{n + a + 1}$

The one idea underneath all twelve derivations

MLE always comes from setting

$$\frac{d}{d\theta} \log p(x \mid \theta) = 0.$$

MAP always comes from setting

$$\frac{d}{d\theta} \left[\log p(x \mid \theta) + \log p(\theta) \right] = 0.$$

The prior only ever contributes “pseudo-counts” (Beta, Gamma, Dirichlet cases) or a “pull toward a prior value” (Gaussian, Laplace, Pareto cases) that get blended with the data-driven answer in exactly the same shape the data itself would have produced with a few extra phantom observations. Ridge regression is this recipe applied to the Gaussian-mean case with $\mu = \theta^\top x^{(i)}$ instead of a scalar constant — the σ^2/τ^2 ratio that appears as λ in ridge is the same precision ratio that appears as the weighting inside every $\hat{\theta}_{\text{MAP}}$ above.